

HIDDEN RADIOLOGY

Chest X-Ray Made Easy

FROM ZERO TO PRO

The Complete Guide to Chest Radiography - Anatomy, Positioning, Quality & Safe Interpretation for
Students & Technicians

By Nasser Mabrouk | Hidden Radiology - ■■■■■■ ■■■■■■

For Educational Purposes Only

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Chapter 1: Introduction to Chest X-Ray

- Chest X-ray is the most common radiological examination worldwide. It is fast, low dose, and provides vital information about lungs, heart, pleura, and bones.
- Indications: cough, chest pain, shortness of breath, trauma, pre-operative check.
- Two standard projections: PA (Postero-Anterior) and Lateral. PA is preferred for better heart magnification control and lower dose to breast and thyroid.
- Key principle: Always compare with old films, check patient name, date, and marker (R/L).

Chapter 2: Anatomy on Chest Film - PA View

- Lungs: Right lung has 3 lobes, left has 2 lobes. Fissures: Horizontal fissure visible on right.
- Heart: Should be less than 50% of thoracic width on PA. Left border: left ventricle. Right border: right atrium.
- Mediastinum: Trachea central, carina at T4-T5 level. Aortic knuckle visible on left.
- Diaphragm: Right hemidiaphragm 1-2 cm higher than left due to liver. Costophrenic angles must be sharp - blunting suggests pleural effusion.
- Bones: Check clavicles, ribs, spine for fractures or lesions. Count ribs - 10 posterior ribs should be visible on good inspiration.

Chapter 3: Lateral View & Additional Projections

- Lateral view: Patient left side against cassette, arms raised, chin up. Useful to localize lesions and see retrocardiac and retrosternal spaces.
- AP view: Used for bedside / ICU patients. Heart appears magnified - do not assess heart size on AP.
- Lordotic view: To see middle lobe and apex. AP with patient leaning back 30-40 degrees.
- Decubitus view: To confirm small pleural effusion - fluid layers.

Chapter 4: Patient Positioning & Breathing Technique

- PA Erect: Patient stands, chest against cassette, hands on hips, elbows forward to rotate scapulae out of lung fields. Chin up.
- SID: 180 cm (72 inch) to reduce magnification.
- Breathing: Deep inspiration, then hold. Good inspiration = 10 posterior ribs visible. Poor inspiration mimics pathology.
- No rotation: Medial ends of clavicles equidistant from vertebral spinous process. Rotation distorts mediastinum.
- For ARDS / ICU patients: Ensure tubes and lines are visible, use grid to reduce scatter.

Chapter 5: Exposure Factors & Image Quality

- kVp: 110-125 kVp high kVp technique for chest to penetrate mediastinum and show lungs behind heart.
- mAs: Low mAs (2-4 mAs) due to low tissue density, but enough to show vascular markings.
- AEC: Use central chamber for PA chest. Ensure proper collimation to include C7 to L3.
- Image Quality Criteria: Sharp vascular markings to periphery, visible retrocardiac vessels, no motion blur, no scapulae over lungs.

Chapter 6: Systematic Approach ABCDE

- A - Airway: Trachea central? Carina? Bronchi?
- B - Breathing: Lungs - compare left vs right, check for pneumothorax, pleural effusion, consolidation, interstitial markings.
- C - Circulation: Heart size, shape, great vessels, hilar vessels.
- D - Diaphragm: Position, shape, costophrenic angles, subdiaphragmatic air?
- E - Everything else: Bones, soft tissues, artifacts, tubes, lines, patient info.
- This systematic approach prevents missing findings.

Chapter 7: Common Technical Errors & How to Avoid

- Rotation: Reposition patient, check clavicles symmetry.
- Poor inspiration: Re-instruct patient, take image on full inspiration.
- Overexposure: Reduce mAs or kVp. Underexposure: Increase.
- Scapulae over lungs: Pull shoulders forward, hands on hips.
- Motion blur: Short exposure time, instruct patient not to breathe.
- Artifacts: Remove necklaces, bras with metal, check for hair braids.

Chapter 8: Radiation Protection in Chest Imaging

- Chest X-ray dose is very low (0.02 mSv). Benefits far outweigh risks.
- Use gonad shielding only if it does not obscure anatomy (not routine for chest).
- Collimate tightly to chest area only.
- Use PA instead of AP when possible to reduce breast dose.
- Pregnancy: Can be done with abdominal shielding if clinically justified. Always ask about pregnancy status.
- ALARA principle: As Low As Reasonably Achievable.

Chapter 9: Checklist & Final Review

- Before exposure: Correct patient ID, correct marker, pregnancy check, remove artifacts, explain breathing.
- After exposure: Check image quality criteria, check for rotation, inspiration, exposure, coverage.
- Documentation: Record exposure factors, any repeat reasons.
- Final Tip: Chest X-ray is pattern recognition skill - see 1000 films to become confident. Use this book as foundation.
- Remember: This book is for educational purposes only. Final diagnosis must be made by qualified radiologist.

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Thank you for learning with us. Next Book: CT Chest Made Easy